



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

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Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. I - Semester							
INFORMATION TECHNOLOGY									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23BS2101	Discrete Mathematics and Graph Theory	BS	3	0	0	3	30	70	100
B23HS2101	Universal Human Values -II : Understanding Harmony and Ethical Human Conduct	HS	2	1	0	3	30	70	100
B23IT2101	Database Management Systems	PC	3	0	0	3	30	70	100
B23IT2102	Object Oriented Programing through Java	PC	3	0	0	3	30	70	100
B23IT2103	Computer Organization	ES	3	0	0	3	30	70	100
B23IT2104	Database Management Systems Lab	PC	0	0	3	1.5	30	70	100
B23IT2105	Object Oriented Programming through JAVA Lab	PC	0	0	3	1.5	30	70	100
B23IT2106	Python Programming	SEC	0	1	2	2	30	70	100
B23MC2101	English Proficiency	MC	2	0	0	-	30	-	30
TOTAL			16	2	8	20	270	560	830

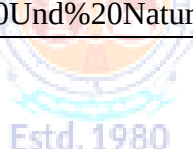
Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2101	BS	3	--	--	3	30	70	3 Hrs.
DISCRETE MATHEMATICS AND GRAPH THEORY								
(Common to CSE, CSBS, AIML, IT, AIDS, CSG, CIC, CSIT)								
Course Objectives: Students are expected to								
1.	Understand the mathematical arguments using logical connectives and quantifiers and verify the validity of arguments using propositional, predicate logic and truth tables.							
2.	Understand various types of relations and discuss various properties of the relations							
3.	Know about the concepts of counting techniques and how to solve the recurrence relations.							
4.	Understand the concepts in graphs and trees.							
Course Outcomes: At the end of the course, Students will be able to								
S.No	Outcome							Knowledge Level
1.	Use the concepts of propositional and predicate logic to verify the arguments for their validity.							K3
2.	Apply the knowledge of set theory to understand relations, functions and their properties.							K3
3.	Solve different counting problems and recurrence relations.							K3
4.	Use the concepts of graphs and their representations.							K3
5.	Determine different multi graphs and tree structures.							K3
SYLLABUS								
UNIT-I (10Hrs)	Mathematical Logic: Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises, Indirect Method of Proof, Predicate Calculus: Predicates, Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.							
UNIT-II (10 Hrs)	Set Theory: Sets: Operations on Sets, Principle of Inclusion-Exclusion, Relations: Properties, Operations, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering, Hasse Diagrams, Lattice and its Properties. Functions: Bijective, Composition, Inverse, Permutation, and Recursive Functions.							
UNIT-III (12Hrs)	Combinatorics and Recurrence Relations: Basis of Counting, Permutations, Permutations with repetitions, Circular and Restricted Permutations, Combinations, Restricted Combinations, Binomial and Multinomial Coefficients and Theorems.							

	Recurrence Relations: Generating Functions, Function of Sequences, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations
UNIT-IV (10 Hrs)	Graph Theory: Basic Concepts, Graph Theory and its Applications, Subgraphs, Graph Representations: Adjacency and Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs.
UNIT-V (08Hrs)	Multi Graphs: Multi graphs, Bipartite and Planar Graphs, Euler's Theorem, Graph Coloring and Covering, Chromatic Number, Trees and their properties, Spanning Trees- BFS and DFS Spanning Trees, Prim's and Kruskal's Algorithms.
Textbooks:	
1.	Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, Tata McGraw Hill.
2.	Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel and T. P. Baker, 2nd Edition, Prentice Hall of India.
Reference Books:	
1.	Elements of Discrete Mathematics-A Computer Oriented Approach, C. L.Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw Hill.
2.	Theory and Problems of Discrete Mathematics, Schaum's Outline Series, Seymour Lipschutz and Marc Lars Lipson, 3rd Edition, McGraw Hill.
3.	Discrete Mathematical Structures, Bernand Kolman, Robert C. Busby and Sharon Cutler Ross, PHI.
4.	Discrete Mathematics, S. K. Chakraborty and B.K. Sarkar, Oxford, 2011.
5.	Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
e-Resources :	
1.	https://nptel.ac.in/courses/106105192
2.	https://archive.nptel.ac.in/courses/111/106/111106102/

Course Code	Category	L	T	P	C	C.I.E	S.E.E	Exam
B23HS2101	HS	2	1	--	3	30	70	3 Hrs.
UNIVERSAL HUMAN VALUES-II: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT								
(Common to all Programmes of Engineering)								
Course Objectives: The objective of this course is to make the student aware of								
1	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.							
2	Harmony in the human being, family, society and nature/existence							
3	Holistic perspective towards life, profession and happiness							
Course Outcomes: At the end of this course student will be able to								
S. No.	Outcome							Knowledge Level
1	Explain the role of value education in achieving basic human aspirations.							K2
2	Summarize needs to obtain harmony in self(I).							K2
3	Describe criteria for human-human relationship and harmony in society							K2
4	Explain four orders of nature and our existence							K2
5	Interpret significance of harmony in holistic development							K2
SYLLABUS								
UNIT-I (9 Hrs)	Introduction to Value Education: Understanding Value Education- Need, Basic Guidelines, Content and Process for Value Education Purpose and motivation for the course. Self-exploration as the Process for Value Education - Sharing about Oneself. Myers-Briggs Type Indicator (MBTI) Personality Test. Continuous Happiness and Prosperity – the Basic Human Aspirations and their Fulfilment. Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Exploring Human Consciousness. Happiness and Prosperity – Current Scenario. Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance-understanding and living in harmony at various levels.							
	Harmony in the Human Being: Understanding Human being as the Co-existence of the Self and the Body. Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self (I) and Body (Happiness and Physical Facility). The Body as an Instrument of the Self (I)' (I being the doer, seer and enjoyer).							

	Understanding Harmony in the Self(I) - Exploring Sources of Imagination in the Self(I). Harmony of the Self (I) with the Body (characteristics and activities of 'I' and harmony in 'I'). Programme to ensure self-regulation(<i>Sanyam</i>) and Health(<i>Swasth</i>)- Exploring Harmony of Self (I) with the Body.
UNIT-III (9 Hrs)	Harmony in the Family and Society: Harmony in the Family – the Basic Unit of Human Interaction. 'Trust' – the Foundational Value in Relationship- Exploring the Feeling of Trust- intention and competence. 'Respect' – as the Right Evaluation- Exploring the Feeling of Respect. Other Feelings, Justice in Human-to-Human Relationship. Understanding Harmony in the Society- (society being an extension of family). Vision for the Universal Human Order- Exploring Systems to fulfil Human Goal.
UNIT-IV (6 Hrs)	Harmony in the Nature/Existence: Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature. Realizing Existence as Co-existence at All Levels - Understanding Existence as Co-existence of mutually interacting units in all pervasive space. Holistic perception of harmony at all levels of existence.
UNIT-V (9 Hrs)	Implications of the Holistic Understanding – a Look at Professional Ethics: Natural acceptance of human values. Definitiveness of (ethical) human conduct. A Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems. Holistic technologies, production systems and management models- typical case studies. Strategies for transition towards value based life and profession (from the present state to Universal Human Order): a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers, b. At the level of society: as mutually enriching institutions and organizations.
Text Books	
1.	R R Gaur, R Sangal, G P Bagaria. "Human Values and Professional Ethics", Excel Books, New Delhi, 2010
2.	R R Gaur, R Asthana, G P Bagaria. "Teachers' Manual for A Foundation Course in Human Values and Professional Ethics", 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2

Reference Books:	
1.	Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).
4.	The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5.	Small is Beautiful - E. F Schumacher
6.	Slow is Beautiful - Cecile Andrews
7.	Economy of Permanence - J C Kumarappa
8.	Bharat Mein Angreji Raj – Pandit Sunderlal
9.	Rediscovering India - by Dharampal
10.	Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11.	India Wins Freedom - Maulana Abdul Kalam Azad
12.	Vivekananda - Romain Rolland (English)
e-Resources	
1.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php
3.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf



ENGINEERING COLLEGE
AUTONOMOUS

Course Code	Category	L	T	P	C	C.I.E	S.E.E	Exam
B23IT2101	PC	3	0	0	3	30	70	3 Hrs.
DATABASE MANAGEMENT SYSTEMS								
(Common to IT, AIDS and CSBS)								
Course Objectives: The main objective of the course is to								
1.	Introduce database management systems and to give a good formal foundation on the relational model of data and usage of Relational Algebra.							
2.	Introduce the concepts of SQL as a universal Database language.							
3.	Demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization							
4.	Provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques							
Course Outcomes: After completion of the course, the student will be able to								
S. No	Outcome							Knowledge Level
1.	Describe database fundamental concepts.							K2
2.	Apply E-R and Relational models for creating databases.							K3
3.	Apply SQL features to create, manipulate and retrieve databases.							K3
4.	Apply normalization concepts to refine relational databases.							K3
5.	Illustrate transaction management concepts.							K2
6.	Apply indexing concepts for searching databases.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database. Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.							
UNIT-II (08 Hrs)	Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus. BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).							

UNIT-III (10 Hrs)	Structured Query Language: Basic SQL Querying Using Select and Where clauses, Arithmetic & Logical Operations, SQL Functions (Date and Time, Numeric to String Conversion). Creating Tables with Relationship, Implementation of Key and Other Integrity Constraints, Set Operations, Nested Queries, Sub Queries, Grouping, Aggregation, Ordering, Implementation of Various Types of Joins, Views (Updatable and Non-Updatable), relational set operations. [CO3, K3]
UNIT-IV (10 Hrs)	Schema Refinement (Normalization): Purpose of Normalization or Schema Refinement, Concept of Functional Dependency, Normal Forms Based on Functional Dependency, Lossless Join and Dependency Preserving Decomposition, (1NF, 2NF and 3 NF), Concept of Surrogate Key, Boyce-Codd Normal Form (BCNF), Multi Valued Dependencies, Fourth Normal Form(4NF), Fifth Normal Form (5NF). [CO4, K3]
UNIT-V (12 Hrs.)	Transaction Management: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm. [CO5, K2] Introduction to Indexing Techniques: B+ Trees, Operations on B+ Trees, Hash Based Indexing: [CO6, K3]
Text Books:	
1.	Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH.
2.	Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH.
Reference Books:	
1.	Introduction to Database Systems, 8th edition, C J Date, Pearson.
2.	Database Management System, 6th edition, Ramez Elmasri, Shamkant B. Navathe, Pearson
3.	Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105175/
2.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2102	PC	3	--	--	3	30	70	3 Hrs.
OBJECT ORIENTED PROGRAMMING THROUGH JAVA								
(Common to IT, AIDS and CSBS)								
Course Objectives: Students are expected to								
1.	To identify Java language components and how they work together in applications.							
2.	To learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.							
3.	To learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications.							
4.	To understand how to design applications with threads in Java.							
5	To understand how to use Java APIs for program development.							
Course Outcomes: By the end of the course, the student will be able to:								
S.No	Outcome							Knowledge Level
1.	Develop simple Java programs that incorporate fundamental programming elements.							K3
2.	Apply the concepts of Object-Oriented Programming such as classes and objects to design java programs							K3
3.	Apply the concepts of inheritance and interfaces to achieve multiple inheritance							K3
4.	Apply the concepts of multithreading and exceptions to design multithreaded error free programs							K3
5.	Model event driven GUI applications which connect with databases.							K4
SYLLABUS								
UNIT-I (10Hrs)	Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style.							
	Input/Output operations: Reading input from console (Scanner).							
	Data Types, Variables, and Operators :Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, Introduction to Operators, Precedence and Associativity of Operators, Assignment Operator, Basic Arithmetic Operators, Increment and Decrement Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators.							
	Control Statements: Introduction, if Expression, Nested if Expressions, if-else							

	Expressions, Ternary Operator, Switch Statement, Iteration Statements, while Expression, do- while Loop, for Loop, Nested for Loop, For-Each for Loop, Break Statement, Continue Statement.
UNIT-II (10 Hrs)	Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Final Class and Methods, Passing Arguments by Value and by Reference, Usage of keyword this. Wrapper classes: Auto boxing and unboxing. Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Nesting of Methods, Attributes Final and Static.
UNIT-III (10 Hrs)	Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three dimensional Arrays, Arrays as Vectors and lists. Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes. Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface, Functional Interfaces, Annotations.
UNIT-IV (10 Hrs)	Packages: Introduction, Defining Package, Importing Packages and Classes into Programs, Access Control, adding a public class and non-public class to an existing user defined package, adding an interface to a user defined package, Different ways of importing packages. Exception Handling: Introduction, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Custom Exceptions, Nested try and catch Blocks. Multithreaded Programming: Introduction, Thread Class, Main Thread- Creation of New Threads, Thread States, Runnable Interface, Thread Priority-Synchronization. Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java
UNIT-V (10 Hrs)	String Handling in Java: Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer. Java Database Connectivity: Introduction, JDBC Architecture, Installing

	MySQL and MySQL Connector/J, JDBC Environment Setup, Establishing JDBC Database Connections, ResultSet Interface. Java FX GUI: Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, laying out nodes in scene graph, mouse events.
Textbooks:	
1.	JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2.	Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3.	JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4 th Edition, Pearson
Reference Books:	
1.	The complete Reference Java, 11th edition, Herbert Schildt, TMH
2.	Murach's Java Programming, Joel Murach
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105191/
2.	https://www.geeksforgeeks.org/java/

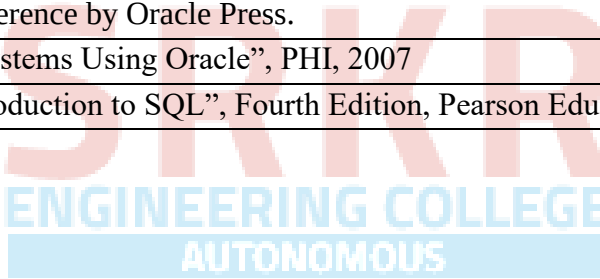
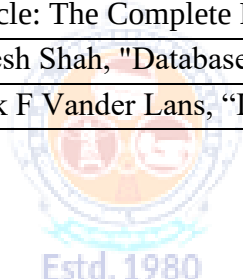


Course Code	Category	L	T	P	C	C.I.E	S.E.E	Exam
B23IT2103	ES	3	--	--	3	30	70	3 Hrs.
COMPUTER ORGANIZATION								
(Common to IT and CSBS)								
Course Objectives: Students are expected to learn								
1.	Principles and the Implementation of Computer Arithmetic							
2.	Operation of CPUs including RTL, ALU, Instruction Cycle, and Busses							
3.	Functionality of central processing unit and control units							
4.	Memory System and I/O Organization							
Course Outcomes: After completion of the course, the student will be able to								
S. No	Outcome							Knowledge Level
1	Identify set of digital components, functional components and micro-operations in a basic computer system.							K3
2	Demonstrate various instructions and arithmetic operations							K3
3	Illustrate knowledge of functional components on central processing unit and various control units.							K2
4	Determine different memory components in a computer for better memory organization							K3
5	Explain different ways of communication with I/O devices and standard I/O interface							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Basic Logic functions, Logic gates, Boolean functions, Canonical forms, Simplification of Boolean functions (up to 4 variable), Basics of Flipflops, Registers, Decoders and multiplexers. Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures. Register Transfer and Micro operations: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit							
UNIT-II (08 Hrs)	Basic Computer Organization and Design: Instruction Codes, Computer Register, Computer Instructions, Instruction Cycle, Memory – Reference Instructions. Input – Output and Interrupt, Complete Computer Description Computer Arithmetic: Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations							

UNIT-III (10 Hrs)	Central Processing Unit: General Register Organization, STACK Organization. Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Execution of a Complete Instruction, Multiple-Bus Organization, Micro programmed Control: Control Memory, Address Sequencing, Micro Program example, Hardwired Control and Micro programmed Control.
UNIT-IV (10 Hrs)	The Memory Organization: Memory Hierarchy, Main memory, Auxiliary memory, Associate Memory, Cache Memory, and Virtual memory, Memory Management Requirements, Secondary Storage.
UNIT-V (12 Hrs.)	Input / Output Organization: Accessing I/O Devices, Interrupts, Processor Examples, modes of transfers, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces.
Text Books:	
1.	Computer System Architecture M. M. Mano:, 3rd ed., Prentice Hall of India, New Delhi, 1993
2.	Digital Design, 6th Edition, M. Morris Mano, Pearson Education.
3.	Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 5/e, McGraw Hill, 2002.
Reference Books:	
1.	Computer Organization and Architecture, William Stallings, 6/e, Pearson, 2006.
2.	Structured Computer Organization, Andrew S. Tanenbaum, 4/e, Pearson, 2005.
3.	Fundamentals of Computer Organization and Design, Sivarama P. Dandamudi, Springer, 2006.
Estd. 1980 AUTONOMOUS	
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105163/
2.	http://www.cuc.ucc.ie/CS1101/David%20Tarnoff.pdf

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2104	PC	--	--	3	1.5	30	70	3 Hrs.
DATABASE MANAGEMENT SYSTEMS LAB								
(Common to IT and CSBS)								
Course Objectives: The major objective of this course is to								
1	Populate and query a database using SQL DDL/DML Commands.							
2	Declare and enforce integrity constraints on a database.							
3	Writing Queries using advanced concepts of SQL.							
4	Programming PL/SQL including procedures, functions, cursors and triggers.							
Course Outcomes: After completing the course, student will be able to:								
S.No	Outcome							Knowledge Level
1	Apply SQL to create and manipulate relational databases using Oracle 11g.							K3
2	Apply SQL to write complex queries.							K3
3	Apply PL/SQL concepts to write triggers, procedures, and programs.							K3
4	Apply SQL concepts to establish database connectivity through high level programs.							K3
SYLLABUS								
1	Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables), examples using SELECT command.							
2	Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.							
3	Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.							
4	Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)							
5	i) Create a simple PL/SQL program which includes declaration section, executable section and exception handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found) ii) Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.							
6	Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.							

7	Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT-IN Exceptions, User defined Exceptions, RAISE_APPLICATION_ERROR.
8	Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES.
9	Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.
10	Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11	Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers.
12	Create a table and perform the search operation on table using indexing and non-indexing techniques.
13	Write a Java program that connects to a database using JDBC.
14	Write a Java program to connect to a database using JDBC and insert values into it.
15	Write a Java program to connect to a database using JDBC and delete values from it.
Reference Books:	
1	Oracle: The Complete Reference by Oracle Press.
2	Nilesh Shah, "Database Systems Using Oracle", PHI, 2007
3	Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2105	PC	--	--	3	1.5	30	70	3 Hrs.
OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB								
(Common to IT and CSBS)								
Course Objectives: The major objective of this course is to								
1	Practice object-oriented programming in the Java programming language.							
2	Implement Classes, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism.							
3	Illustrate inheritance, Exception handling mechanism, JDBC connectivity.							
4	Construct Threads, Event Handling, implement packages, Java FX GUI.							
Course Outcomes: By the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1	Apply control statements, operators, strings and arrays to design java programs							K3
2	Apply Object Oriented Programming concepts to design java programs							K3
3	Design multithreaded and error free java programs by applying the concepts of Exception Handling and Multithreading							K4
4	Design interactive GUI programs by applying the concepts of JAVAFX and JDBC							K4
SYLLABUS								
1	a) Write a JAVA program to display default value of all primitive data type of JAVA b) Write a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on value of D, describe the nature of root.							
2	a) Write a JAVA program to search for an element in a given list of elements using binary search mechanism. b) Write a JAVA program to sort for an element in a given list of elements using bubble sort c) Write a JAVA program using StringBuffer to delete, remove character.							
3	a) Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method. b) Write a JAVA program implements method overloading. c) Write a JAVA program to implement constructor. d) Write a JAVA program to implement constructor overloading.							
4	a) Write a JAVA program to implement Single Inheritance b) Write a JAVA program to implement multi-level Inheritance c) Write a JAVA program for abstract class to find areas of different shapes							

5	a) Write a JAVA program give example for “super” keyword. b) Write a JAVA program to implement Interface. What kind of Inheritance can be achieved? c) Write a JAVA program that implements Runtime polymorphism
6	a) Write a JAVA program that describes exception handling mechanism b) Write a JAVA program Illustrating Multiple catch clauses c) Write a JAVA program for creation of Java Built-in Exceptions d) Write a JAVA program for creation of User Defined Exception
7	a) Write a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds, (Repeat the same by implementing Runnable) b) Write a program illustrating is Alive and join () c) Write a Program illustrating Daemon Threads.
8	a) Write a JAVA program that import and use the user defined packages b) Without writing any code, build a GUI that display text in label and image in an ImageView (use JavaFX) c) Build a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI
9	a) Write a java program that connects to a database using JDBC b) Write a java program to connect to a database using JDBC and insert values into it. c) Write a java program to connect to a database using JDBC and delete values from it
Reference Books:	
1	The complete Reference Java, 11th edition, Herbert Schildt, TMH
2	Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
23BIT2106	SEC	--	1	2	2	30	70	3 Hrs.
PYTHON PROGRAMMING								
(For IT)								
Course Objectives: The major objective of this course is to								
1	Introduce core programming concepts of Python programming language							
2	Demonstrate about Python data structures like Lists, Tuples, Sets and dictionaries							
3	Implement Functions, Modules and Regular Expressions in Python Programming and to create practical and contemporary applications using these							
Course Outcomes: After completion of the course student will be able to								
S.No	Outcome							Knowledge Level
1	Demonstrate Basic Python Programming Concepts for solving problems							K3
2	Analyze Python Functions and Modules for real time applications							K4
3	Evaluate data collection concepts on Data Handling.							K4
4	Make use of NumPy, Pandas for data preprocessing.							K3
SYLLABUS								
Unit-I	History of Python Programming Language, Thrust Areas of Python, Installing Anaconda Python Distribution, Installing and Using Jupyter Notebook. Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language. Control Flow Statements: if statement, if-else statement, if...elif...else, Nested if statement, while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.							
	Sample Experiments:							
1	Write a program to find the largest element among three Numbers.							
2	Write a Program to display all prime numbers within an interval							
3	Write a program to swap two numbers without using a temporary variable.							
4	Demonstrate the following Operators in Python with suitable examples. i) Arithmetic Operators ii) Relational Operators iii) Assignment Operators iv) Logical Operators v) Bit wise Operators vi) Ternary Operator vii) Membership Operators viii) Identity Operators							
5	Write a program to add and multiply complex numbers							
6	Write a program to print multiplication table of a given number							

Unit-II	Functions: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments. Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings. Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, del Statement
	Sample Experiments:
7	Write a program to define a function with multiple return values.
8	Write a program to define a function using default arguments.
9	Write a program to find the length of the string without using any library functions.
10	Write a program to check if the substring is present in a given string or not.
11	Write a program to perform the given operations on a list: i. Addition ii. Insertion iii. slicing
12	Write a program to perform any 5 built-in functions by taking any list.
Unit-III	Dictionaries: Creating Dictionary, Accessing and Modifying key:value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, del Statement. Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset.
	Sample Experiments:
13	Write a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples
14	Write a program to count the number of vowels in a string (No control flow allowed).
15	Write a program to check if a given key exists in a dictionary or not
16	Write a program to add a new key-value pair to an existing dictionary
17	Write a program to sum all the items in a given dictionary
Unit-IV	Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules. Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism.
	Sample Experiments:
18	Write a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered
19	Python program to print each line of a file in reverse order.
20	Python program to compute the number of characters, words and lines in a file.
21	Write a program to create, display, append, insert and reverse the order of the items in the array
22	Write a program to add, transpose and multiply two matrices.

23	Write a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.
Unit-V	
Introduction to Data Science: Functional Programming, JSON and XML in Python, NumPy with Python, Pandas.	
Sample Experiments:	
24	Python program to check whether a JSON string contains complex object or not.
25	Python Program to demonstrate NumPy arrays creation using array () function.
26	Python program to demonstrate use of ndim, shape, size, dtype
27	Python program to demonstrate basic slicing, integer and Boolean indexing
28	Python program to find min, max, sum, cumulative sum of array
29	Create a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data frame as follows: a) Apply head () function to the pandas data frame b) Perform various data selection operations on Data Frame
30	Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib
Reference Books:	
1	Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2	Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2 nd Edition, Pearson, 2024
3	Introduction to Programming Using Python, Y. Daniel Liang, Pearson

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2101	MC	2	--	--	--	30	--	--
ENGLISH PROFICIENCY								
(Common to ME, IT, CSIT, AIDS& AIML)								
Course Outcomes: After completion of this course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Interpret IELTS & TOEFL listening comprehension texts.							K2
2.	Demonstrate essential speaking skills in academic, professional, and real-life contexts.							K2
3.	Interpret the written discourse by applying effective reading strategies.							K2
4.	Construct coherent and cohesive paragraphs, e-mails, letters, and essays.							K3
SYLLABUS								
UNIT-I	Listening Skills Listening as a receptive skill Listening skills for IELTS, TOEFL, Duolingo, PTE, and other competitive examinations.							
UNIT-II	Speaking Skills JAM/ Extempore Debate / Group Discussion Presentation Skills							
UNIT-III	Reading Skills Types of Reading(Intensive and Extensive Reading, Skimming, Scanning) Reading/Summarizing News Paper Articles							
UNIT-IV	Writing Skills Essay Writing(Argumentative, Analytical and Descriptive essays for GRE, TOEFL & IELTS) E-Mail / Letter Writing Resume Writing							
UNIT-V	Integrated Language Skills Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking							
Text Books:								
1.	Interchange (5 th edition) by Jack C.Richards, CUP, 2017.							
Reference Books:								

1.	Fundamentals of Technical Communication (1 st edition) by Meenakshi Raman, Sangeeta Sharma of OUP, 2014.
2.	The Oxford Guide to Writing and Speaking (3 rd edition) by John Seely OUP, 2013.
3.	Effective Technical Communication (2 nd edition) by M. Ashraf Rizwi. TataMcGrawhill, 2017.
e-Resources:	
1	BBC Learning English - Learn English with BBC Learning English - Homepage
2	Grammar Learn English (britishcouncil.org)
3	Duolingo English Test
4	IELTS Test Preparation Materials - Videos, Practice tests, Articles and More (idp.com)





SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. II - Semester							
INFORMATION TECHNOLOGY									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23HS2203	Optimization Techniques	HS	2	0	0	2	30	70	100
B23BS2201	Probability & Statistics	ES	3	0	0	3	30	70	100
B23IT2201	Advanced Data Structures & Algorithms	PC	3	0	0	3	30	70	100
B23IT2202	Software Engineering	PC	3	0	0	3	30	70	100
B23IT2203	Operating Systems	PC	3	0	0	3	30	70	100
B23IT2204	Operating Systems & Software Engineering Lab	PC	0	0	3	1.5	30	70	100
B23IT2205	Advanced Data Structures Lab	PC	0	0	3	1.5	30	70	100
B23IT2206	Python with Django	SEC	0	1	2	2	30	70	100
B23IT2207	Design Thinking & Innovation	ES	1	0	2	2	30	70	100
B23MC2202	Environmental Science	MC	2	0	0	-	30	-	30
TOTAL			17	1	10	21	300	630	930

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2203	HS	2	--	--	2	30	70	3 Hrs.
OPTIMIZATION TECHNIQUES								
(Common to AIML & IT)								
Course Objectives:								
1.	To define an objective function and constraint functions in terms of design variables, and then state the optimization problem.							
2.	To state single variable and multi variable optimization problems, without and with constraints and solve by classical methods.							
3.	To apply linear programming technique to an optimization problem, define slack and surplus variables, by using Simplex method.							
4.	To apply transportation technique to real life problems of transportation.							
5.	To solve Integer programming problems that encounter in real life practical cases.							
6.	To design optimum schedules for projects.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply classical optimization techniques to minimize or maximize single and multi-variable optimization problems without or with constraints and arrive at an optimal solution.							K3
2.	Apply suitable optimization algorithms to formulate and solve linear programming problems.							K3
3.	Apply specialized methods to determine optimal schedules for transportation problems.							K3
4.	Solve Integer programming problems by using suitable optimization algorithms.							K3
5.	Apply analytical methods to project networks to determine optimal time schedules.							K3
SYLLABUS								
UNIT-I (10Hrs)	Classical Optimization Techniques: Statement of an Optimization problem, classification of Optimization problems, Single variable optimization without and with constraints, multi variable optimization without constraints, multivariable optimization with equality constraints - solution by the method of Lagrange multipliers, multivariable optimization with inequality constraints – Kuhn Tucker conditions.							

UNIT-II (10 Hrs)	Linear Programming: Definition and formulation of LPP, Graphical solution, Standard form of a linear programming problem, Simplex method and Algorithm, Artificial Variable Technique – Big M method.
UNIT-III (10 Hrs)	Transportation Problem: Mathematical Formulation, Tabular Representation, Balanced and unbalanced transportation Problems - Initial Solution by north – west corner rule, least cost method and Vogel’s approximation method, testing for optimality by MODI method.
UNIT-IV (10 Hrs)	Integer Programming: Introduction to Integer programming, methods of integer programming, Gomory’s cutting plane method - All Integer Programming Problems, Branch and Bound method for all integer programming problems.
UNIT-V (10 Hrs)	Network Analysis: Introduction, Project Scheduling by CPM and PERT, Network diagram representation, rules for drawing network diagram, Labelling by Fulkerson’s rule, Network calculations - EST, EFT, LST, LFT, Float/Slack and critical path, CPM and PERT calculations.
Textbooks:	
1.	“Engineering optimization: Theory and practice”, S. S.Rao, New Age International (P) Limited, 3rd edition, 1998.
2.	Operations Research, Dr. S. D. Sharma, Kedarnath, Ramnath & Co
Reference Books:	
1.	“Optimization Methods in Operations Research and systems Analysis”, by K.V. Mital and C. Mohan, New Age International (P) Limited, Publishers, 3rd edition, 1996.
2.	“Introductory Operations Research”, H.S. Kasene & K.D. Kumar, Springer (India), Pvt.LTd
e-Resources	
1.	https://nptel.ac.in/courses/112/106/112106134/
2.	https://nptel.ac.in/courses/110/106/110106062/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2201	ES	3	0	0	3	30	70	3 Hrs.
PROBABILITY AND STATISTICS								
(Common to AIML, CSE, CSBS, CSIT and IT)								
Course Objectives: Students are expected to								
1.	Acquire critical thinking skills from the concepts of Descriptive Statistics, Data Science and Probability.							
2.	Predict the relationship and the impact of the relationship between the variables.							
3.	Obtain Decision-making skills from Sampling theory.							
Course Outcomes: Upon the successful completion of this course, Students will								
S.No.	Outcome							Knowledge Level
1.	Compute various statistical measures like central tendency and spread values.							K2
2.	Use the concepts of probability and random variables to solve simple problems based on discrete and continuous probability distributions.							K3
3.	Determine correlation and regression coefficients and model a best suitable curve for a given data using the method of least squares.							K3
4.	Apply the procedures of sampling theory to find point and interval estimates for various sampling distributions.							K3
5.	Model a framework by testing of hypothesis for getting inferences about Population Parameters based on Sample statistic.							K3
SYLLABUS								
UNIT-I (10 Hrs.)		Descriptive statistics and methods for data science: Data science- Statistics Introduction- Population vs Sample –Collection of data: primary and secondary data- Type of variables: dependent and independent, Categorical and Continuous variables- Data visualization- Measures of Central tendency- Measures of Variability (spread or variance)- Moments- Measures of Skewness and Kurtosis.						
UNIT-II (12 Hrs.)		Probability, Random variables and Distributions: Probability– Conditional probability,addition, Multiplication theorems and Baye’s theorem. Random variables – Discrete and Continuous random variables – Distribution functions – Probability mass function, Probability density function and Cumulative distribution functions – Mathematical Expectation – Variance. Discrete and Continuous Distributions: Discrete Distributions: Bernoulli, Binomial and Poisson distributions - Mean, Variance, Fitting of distributions.						

	Continuous Distributions: Uniform distribution, Normal Distribution, Standard Normal Variate - Mean, Variance.
UNIT-III (12 Hrs.)	Correlation, Regression and Curve fitting: Correlation: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's rank correlation coefficient (without proofs). Linear Regression: Regression- Regression coefficients and properties–Multiple Linear Regression. Curve Fitting: Method of Least Squares, Fitting of a Straight line, Fitting of a Parabola, Fitting of an Exponential curves: $y = ae^{bx}$, $y = ab^x$ and Power curve: $y = ax^b$.
UNIT-IV (8 Hrs.)	Sampling Theory: Introduction – Population and Samples –Parameter and Statistic- Sampling distribution of statistic-Standard error- Sampling distribution of Means and Variance (definition only) – Point and Interval estimations. Maximum error of estimate – Central limit theorem (without proof) – Estimation using t, χ^2 and F-distributions.
UNIT-V (12 Hrs.)	Testing of Hypothesis: Testing of Hypothesis- Formulation of Null hypothesis, Alternative hypothesis, Critical region, level of significance, Errors in sampling- Type-I-error, Type-II-error, One-tailed and Two-tailed tests, Degrees of freedom. Large Sample Theory: Test of significance for single and difference of Proportions. Small Sample Theory: Student's-t-distribution: Definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means. F-distribution: Definition, F-test for equality of two population variances. Chi-square distribution: Definition, Chi-square test for attributes and goodness of fit.
Text Books:	
1.	Fundamentals of Mathematical Statistics by S. C. Gupta and V.K. Kapoor, Sultan Chand & Sons Publishers.
2	Probability and Statistics for Engineers, Miller and Freund, 7 th edition, Prentice-Hall India.
Reference Books:	
1.	Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Eighth edition, Pearson Education.
2.	Probability and Statistics for Engineering and the Sciences, Jay I. Devore, 8 th Edition, Cengage.
3.	Introduction to probability and statistics Engineers and the Scientists, Sheldon M. Ross, 4 th Edition, Academic Foundation, 2011.
4.	Johannes Ledolter and Robert V. Hogg, Applied statistics for Engineers and Physical

	Scientists, 3rd Edition, Pearson, 2010.
5.	Probability, Statistics and Random Processes by T.Veerarajan, Tata McGraw Hill Pub.
6.	Higher Engineering Mathematics, by Dr.B.S.Grewal, 43 rd Edition, Khanna Publishers
7.	Paul L. Meyer, Introductory Probability and Statistical Applications (2 nd edn.), Addison-Wesley, 1970.
e-Resources:	
1.	http://www.swayam.gov.in
2.	https://archive.nptel.ac.in/courses/106/104/106104233/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2201	PC	3	--	--	3	30	70	3 Hrs.
ADVANCED DATA STRUCTURES & ALGORITHMS								
(For IT)								
Course Objectives: The major objective of this course is to								
1.	Provide knowledge on advance data structures frequently used in Computer Science domain							
2.	Develop skills in algorithm design techniques popularly used							
3.	Understand the use of various data structures in the algorithm design							
Course Outcomes: After completion of the course student will be able to								
S.No	Outcome							Knowledge Level
1.	Apply non-linear data structures for solving a given problem.							K3
2.	Apply divide and conquer technique to solve problems and calculate time complexity.							K3
3.	Apply different design technique algorithms like greedy, dynamic programming, backtracking and branch and bound for solving problems.							K3
4.	Apply the knowledge of complexity classes P, NP and NP-Complete for solving graph and Scheduling problems							K3
Estd. 1980 SYLLABUS								
UNIT-I (10Hrs)	AVL Trees – Creation, Insertion, Deletion operations and Applications B-Trees – Creation, Insertion, Deletion operations and Applications Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications							
UNIT-II (10 Hrs)	Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations. Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen’s matrix multiplication, Convex Hull							
UNIT-III (10 Hrs)	Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths– General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem							

UNIT-IV (10 Hrs)	Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem
UNIT-V (10 Hrs)	NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem NP Hard Graph Problems: Clique Decision Problem (CDP), Chromatic Number Decision Problem (CNDP), Traveling Salesperson Decision Problem (TSP) NP Hard Scheduling Problems: Scheduling Identical Processors, Job Shop Scheduling
Textbooks:	
1.	Fundamentals of Data Structures in C++, Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh, 2 nd Edition Universities Press
2.	Computer Algorithms in C++, Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, 2 nd Edition University Press
Reference Books:	
1.	Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2.	An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
3.	The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4.	Data Structures using C & C++: Langsam, Augenstein & Tanenbaum, Pearson, 1995
5.	Algorithms + Data Structures & Programs:, N.Wirth, PHI
6.	Fundamentals of Data Structures in C++: Horowitz Sahni & Mehta, Galgottia Pub.
7.	Data structures in Java:, Thomas Standish, Pearson Education Asia
e-Resources	
1.	https://www.tutorialspoint.com/advanced_data_structures/index.asp
2.	http://peterindia.net/Algorithms.html
3.	Abdul Bari, Introduction to Algorithms (youtube.com)

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2202	PC	3	--	--	3	30	70	3 Hrs.
SOFTWARE ENGINEERING								
(Common to IT and CSBS)								
Course Objectives: The major objective of this course is to								
1.	Exposure to different phases of software development, including Waterfall, Unified Process, and agile process models.							
2.	Hands-on experience with various software engineering practices, such as requirements analysis, code analysis, debugging, testing, traceability, and version control.							
3.	Exposure to software design techniques.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply software engineering concepts to illustrate various phases of a lifecycle and choose appropriate process model depending on the user requirements.							K3
2.	Apply software project management concepts to prepare software requirements specification report.							K3
3.	Apply various software architectures and design-patterns to select the appropriate techniques to design the software and User Interface.							K3
4.	Examine whether all the requirements specified have been achieved or not under specified quality standards.							K3
5.	Apply Computer-Aided Software Engineering (CASE) tools for managing projects.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Evolution, Software development projects, Exploratory style of software developments, Emergence of software engineering, Notable changes in software development practices, Computer system engineering. Software Life Cycle Models: Basic concepts, Waterfall model and its extensions, Rapid application development, Agile development model, Spiral model.							
UNIT-II (10 Hrs)	Software Project Management: Software project management complexities, Responsibilities of a software project manager, Metrics for project size estimation, Project estimation techniques, Empirical Estimation techniques, COCOMO, Halstead’s software science, risk management. Requirements Analysis And Specification: Requirements gathering and analysis, Software Requirements Specification (SRS), Formal system specification, Axiomatic specification, Algebraic specification, Executable specification and 4GL.							

UNIT-III (10 Hrs)	Software Design: Overview of the design process, how to characterize a good software design? Layered arrangement of modules, Cohesion and Coupling approaches to software design. Agility: Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models. (Text Book 2) Function-Oriented Software Design: Overview of SA/SD methodology, Structured analysis, Developing the DFD model of a system, Structured design, Detailed design, and Design Review. User Interface Design: Characteristics of a good user interface, Basic concepts, Types of user interfaces, Fundamentals of component-based GUI development, and user interface design methodology.
UNIT-IV (10 Hrs)	Coding and Testing: Coding, Code review, Software documentation, Testing, Black-box testing, White-Box testing, Debugging, Program analysis tools, Integration testing, testing object-oriented programs, Smoke testing, and Some general issues associated with testing. Software Reliability and Quality Management: Software reliability. Statistical testing, Software quality, Software quality management system, ISO 9000.SEI Capability maturity model. Few other important quality standards, and Six Sigma.
UNIT-V (10 Hrs)	Computer-Aided Software Engineering (Case): CASE and its scope, CASE environment, CASE support in the software life cycle, other characteristics of CASE tools, Towards second generation CASE Tool, and Architecture of a CASE Environment. Software Maintenance: Characteristics of software maintenance, Software reverse engineering, Software maintenance process models and Estimation of maintenance cost. Software Reuse: reuse- definition, introduction, reason behind no reuse so far, Basic issues in any reuse program, A reuse approach, and Reuse at organization level.
Textbooks:	
1.	Fundamentals of Software Engineering, Rajib Mall, 5 th Edition, PHI.
2.	Software Engineering A practitioner's Approach, Roger S. Pressman, 9 th Edition, Mc-Graw Hill International Edition.
Reference Books:	
1.	Software Engineering, Ian Sommerville, 10 th Edition, Pearson.
2.	Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105182/
2.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview
3.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013382690411003904735_shared/overview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2203	PC	3	--	--	3	30	70	3 Hrs.
OPERATING SYSTEMS								
(Common to IT and CSBS)								
Course Objectives: The major objective of this course is to								
1.	Understand the basic concepts and principles of operating systems, including process management, memory management, file systems, and Protection.							
2.	Make use of process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.							
3.	Illustrate different conditions for deadlock and their possible solutions.							
Course Outcomes: After completion of the course student will be able to								
S.No	Outcome							Knowledge Level
1.	Describe Operating System features and system structures.							K2
2.	Apply CPU Scheduling Algorithms for Multi Process and Multi Threaded Operating systems.							K3
3.	Solve Process Synchronization problems to avoid occurrence of Deadlock situations							K3
4.	Apply various Memory Management Schemes for Primary and Secondary memory.							K3
5.	Describe file Operations and protection methods.							K2
SYLLABUS								
UNIT-I (10Hrs)	Operating Systems Overview: Introduction, Operating system functions, Operating systems operations, Computing environments, Free and Open-Source Operating Systems System Structures: Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, Operating system Design and Implementation, Operating system structure, Building and Booting an Operating System, Operating system debugging.							
UNIT-II (10 Hrs)	Processes: Process Concept, Process scheduling, Operations on processes, Inter-process communication. Threads and Concurrency: Multithreading models, Thread libraries, Threading issues. CPU Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling.							

UNIT-III (10 Hrs)	Synchronization Tools: The Critical Section Problem, Peterson's Solution, Mutex Locks, Semaphores, Monitors, Classic problems of Synchronization. Deadlocks: system Model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlock.
UNIT-IV (10 Hrs)	Memory-Management Strategies: Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping. Virtual Memory Management: Introduction, Demand paging, Copy-on-write, Page replacement, Allocation of frames, Thrashing. Storage Management: Overview of Mass Storage Structure, HDD Scheduling.
UNIT-V (10 Hrs)	File System: File System Interface: File concept, Access methods, Directory Structure. File system Implementation: File-system structure, File-system Operations, Directory implementation, Allocation method, Free space management. File-System Internals: FileSystem Mounting, Partitions and Mounting, File Sharing. Protection: Goals of protection, Principles of protection, Protection Rings, Domain of protection, Access matrix.
Textbooks:	
1.	Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition Wiley, 2018.
2.	Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson , 2016.
Reference Books:	
1.	Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018.
2.	Operating Systems: A Concept Based Approach, D.M Dhamdhare, 3rd Edition, McGraw- Hill, 2013.
3.	Nutt G, Operating Systems, 3rd edition, Pearson Education, 2004.
e-Resources	
1.	https://nptel.ac.in/courses/106106144
2.	https://peterindia.net/OperatingSystems.html

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2204	PC	--	--	3	1.5	30	70	3 Hrs.
OPERATING SYSTEMS & SOFTWARE ENGINEERING LAB								
(Common to IT and CSBS)								
Course Objectives: The major objective of this course is to								
1	Provide insights into system calls, file systems, semaphores.							
2	Develop and debug CPU Scheduling algorithms, page replacement algorithms, thread implementation							
3	Implement Bankers Algorithms to Avoid the Dead Lock							
4	acquire the generic software development skill through various stages of software life cycle							
5	generate test cases for software testing							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1	Make use of Unix commands and System calls to explore the operating system functionalities.							K3
2	Apply various Operating System services like process scheduling & synchronization, memory management, deadlock handling and File management.							K3
3	Apply Object oriented concepts for Software Development Life Cycle using Rational Rose.							K3
4	Design UML diagrams for any real world application.							K4
SYLLABUS								
Experiments covering the Topics:								
	• UNIX fundamentals, commands & system calls • CPU Scheduling algorithms, thread processing • IPC, semaphores, monitors, deadlocks • Page replacement algorithms, file allocation strategies • Memory allocation strategies • Software Requirement Specification, DFD, CFD • Software estimation, UML diagrams, test case design							
Sample Experiments in Operating Systems:								
1	Practicing of Basic UNIX Commands(ex: cat, touch, cp, ls, mv, rm, mkdir, rmdir, grep, sort, cut, wc, chmod, sed, awk)							
2	Write programs using the following UNIX operating system calls fork, vfork, exec, getpid, wait, close, read and write							

3	Simulate UNIX commands like cp, ls and grep
4	Simulate the following CPU scheduling algorithms a) FCFS b) SJF c) Round Robin
5	Write a program to illustrate concurrent execution of threads using pthreads library.
6	Write a program to solve producer-consumer problem using Semaphores.
7	Implement Bankers Algorithm for Dead Lock avoidance
8	Implement the following memory allocation methods a) MFT b) MVT
9	Simulate the following page replacement algorithms a) FIFO b) LRU
10	Simulate the following file allocation strategies a) Sequential b) Indexed c) Linked
11	Download and install nachos operating system and experiment with it

Sample Experiments in Software Engineering:

1	Perform the following, for the following experiments: I. Do the Requirement Analysis and Prepare SRS II. Draw E-R diagrams, DFD, CFD and structured charts for the project. a. Course Registration System b. Students Marks Analyzing System c. Online Ticket Reservation System d. Stock Maintenance
2	Consider any application, using COCOMO model, estimate the effort.
3	Consider any application, Calculate effort using FP oriented estimation model.
4	Draw the UML Diagrams for the problem a, b, c, d.
5	Design the test cases for e-Commerce application (Flipkart, Amazon)
6	Design the test cases for a Mobile Application (Consider any example from App store)
7	Design and Implement ATM system through UML Diagrams.

Reference Books:

1	“Unix Shell Programming” by Yashwant P kanitkar, BPB Publications.
2	“Introduction to UNIX & SHELL programming”, by M.G. Venkatesh Murthy, Pearson Education.
3	“Operating System Concepts”, Silberschatz A, Galvin P B, and Gagne G, 9th edition, Wiley, 2013.
4	“Let Us C: Authentic guide to C programming language“, by Yashavant Kanetkar, 19th Edition
5	“Unified Modeling Language User Guide” by Booch, Pearson Education.
6	Object Oriented Software Engineering using UML, Patrons and Java by Bernd Bruegge, Allen H. Dutoit, 3rd Edition, Prentice Hall Press

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2205	PC	--	--	3	1.5	30	70	3 Hrs.
ADVANCED DATA STRUCTURES LAB								
(For IT)								
Course Objectives: The major objective of this course is to								
1	Acquire practical skills in constructing and managing Data structures							
2	Apply the popular algorithm design methods in problem-solving scenarios							
Course Outcomes: After completion of the course student will be able to								
S. No	Outcome							Knowledge Level
1	Apply Non-Linear data structures for solving any given problem.							K3
2	Apply divide and conquer technique for solving quick and merge sort problems.							K3
3	Apply greedy method for solving Job sequencing and Single source shortest path problems.							K3
4	Use dynamic programming, Backtracking and branch and bound for solving given problems.							K3
SYLLABUS								
1	Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.							
2	Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.							
3	Construct Min and Max Heap using arrays, delete any element and display the content of the Heap							
4	Implement BFT and DFT for given graph, when graph is represented by a) Adjacency Matrix b) Adjacency Lists							
5	Write a program for finding the biconnected components in a given graph.							
6	Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst and Best cases).							
7	Compare the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.							
8	Implement Job Sequencing with deadlines using Greedy strategy.							
9	Write a program to solve 0/1 Knapsack problem Using Dynamic Programming.							
10	Implement N-Queens Problem Using Backtracking.							
11	Use Backtracking strategy to solve 0/1 Knapsack problem.							
12	Implement Travelling Salesperson problem using Branch and Bound approach.							

Reference Books:	
1	Fundamentals of Data Structures in C++, Horowitz Ellis, Sahni Sartaj, Mehta, Dinesh, 2nd Edition, Universities Press
2	Computer Algorithms/C++ Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, 2nd Edition, University Press
3	Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4	An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2206	SEC	--	1	2	2	30	70	3 Hrs.
PYTHON WITH DJANGO								
(For IT)								
Course Objectives: The major objective of this course is to								
1	Design and build static as well as dynamic web pages and interactive web-based applications							
2	Web development using Django framework.							
3	Analyze and create functional website in Django and deploy Django Web Application on Cloud							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1	Apply static web pages using HTML, CSS, and potentially JavaScript							K2
2	Apply Django framework for developing dynamic web applications.							K3
3	Apply Django framework on SQLite for developing web applications and deploy in cloud.							K3
SYLLABUS								
UNIT-I	Python libraries for web development : Collections-Container datatypes, Tkinter-GUI applications, Requests-HTTP requests, BeautifulSoup4-web scraping, Scrapy, Zappa, Dash, CherryPy, Turbo Gears, Flask, Web2Py, Bottle, Falcon, Cubic Web, Quixote, Pyramid							
1	Sample Experiments: Write a Python GUI program to import Tkinter package and create a window. Set its title and add a label to the window.							
2	Write a Python program that designs a simple login form with labels and Entry widgets, arranging them in a grid using the Grid geometry manager.							
3	Write a program using BeautifulSoup4 library for web scraping for a given URL							
4	Develop a sample Hello World page using Flask framework							
5	Develop a sample web page using CherryPy / Web2Py / Bottle Framework							
UNIT-II	Introduction to Django Framework Understanding Django environment, Features of Django and Django architecture, MVC and MTV, Urls and Views, Mapping the views to URLs, Django Template, Template inheritance Django Models, Creating model for site, Converting the model into a table, Fields in Models, Integrating Bootstrap into Django, Creating tables, Creating grids, Creating carousels.							

6	Sample Experiments: Create a Sample “Hello World” Application using Django
7	Create a Login and Registration Page using MVC architecture in Django Framework
8	Create a sample page in Django by integrating BootStrap.
9	Create an application with Tables, grids in Django
10	Create a Django App with Carousels feature.
UNIT-III	Integrating Accounts & Authentication on Django Introduction to Django Authentication System, Security Problem &Solution with Django Creating Registration Form using Django, Adding Email Field in Forms, Configuring email settings, Sending emails with Django, Adding Grid Layout On Registration Page, Adding Page Restrictions, Login Functionality Test and Logout.
11	Sample Experiments: Create a registration page using Authentication System
12	Create an application in Django to send emails using email settings and Grid Layout
13	Create an application in Django using page restriction / authentication with Login and Logout Functionality
14	Create a sample form using Django Forms
UNIT-IV	Connecting SQLite with Django Database Migrations, Fetch Data from Database, Displaying Data On Templates, Adding Condition On Data, Sending data from url to view, Sending data from view to template, Saving objects into database, Sorting objects, Filtering objects, Deleting objects, Difference between session and cookie, Creating sessions and cookies in Django.
15	Sample Experiments: Create an app in Django which fetches data from database and show as list and also save objects in database
16	Create an app in Django for performing CRUD operations on records in a database
17	Create an app in Django which uses session management and cookies to store and manage user sessions.
UNIT-V	Deploying Django Web Application on Cloud Creating a functional website in Django, Four Important Pillars to Deploy, registering on Heroku and GitHub, Push project from Local System to GitHub, working with Django Heroku, Working with Static Root, Handling WSGI with gunicorn, setting up Database & adding users.
18	Sample Experiments: Create a website in Django with login, and registration page.
19	Register on GitHub, and Heroku and deploy the website on Heroku with all the functionalities developed.
20	Configure Django to handle static files.

Text books:	
1	Martin C.Brown, “Python: The Complete Reference Paper back”, 4 th Edition 2018, McGraw Hill Education
2	Reema Thareja, “Python Programming: Using Problem Solving Approach”, 3 rd Edition 2017, Oxford.
3	Daniel Rubio, Apress,” Beginning Django Web Application Development and Deployment with Python”, 2 nd Edition 2017, Apress
Reference Books:	
1	Tom Aratyn, “Building Django 2.0 Web Applications: Create enterprise-grade, scalable Python web applications easily with Django 2.0”,2 nd Edition 2018, Packt Publishing
2	Harry Percival, “Test-Driven Development with Python: Obey the Testing Goat: Using Django, Selenium and JavaScript”,2 nd Edition 2019, Kindle Edition.



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT2207	ES	1	--	2	2	30	70	3 Hrs.
DESIGN THINKING & INNOVATION								
(Common to all Programmes of Engineering)								
Course Objectives:								
1.	Bring awareness on innovative design and new product development.							
2.	Explain the basics of design thinking.							
3.	Familiarize the role of reverse engineering in product development.							
4.	Train how to identify the needs of society and convert into demand.							
5.	Introduce product planning and product development process.							
Course Outcomes: After completion of this course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Define the concepts related to design thinking.							K1
2.	Explain the fundamentals of Design Thinking and innovation.							K2
3.	Apply the design thinking techniques for solving problems in various sectors.							K3
4.	Analyse to work in a multidisciplinary environment.							K4
5.	Evaluate the value of creativity.							K5
SYLLABUS								
UNIT-I (10Hrs)	Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.							
UNIT-II (10 Hrs)	Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development. Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.							
UNIT-III (10 Hrs)	Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations. Creativity to Innovation. Teams for innovation, Measuring the impact and value of creativity. Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.							
UNIT-IV (10 Hrs)	Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications. Innovation towards product design Case studies.							

	Activity: Importance of modeling, how to set specifications, Explaining their own product design.
UNIT-V (10 Hrs)	Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business Cases. Developing & testing prototypes. Activity: How to market our own product, about maintenance, Reliability and plan for startup.
Textbooks:	
1.	Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2.	Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.
Reference Books:	
1.	David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2.	Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3.	William lidwell, Kritinaholden, &Jill butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4.	Chesbrough.H, The era of open innovation, 2003.
e-Resources:	
1.	https://nptel.ac.in/courses/110/106/110106124/
2.	https://nptel.ac.in/courses/109/104/109104109/
3.	https://swayam.gov.in/nd1_noc19_mg60/preview
4.	https://onlinecourses.nptel.ac.in/noc22_de16/preview

Course Code	Category	L	T	P	C	C.I.E	S.E.E	Exam
B23MC2202	MC	2	--	--	--	30	--	--
ENVIRONMENTAL SCIENCE								
(Common to AIDS, AIML, CE, CSBS, CSIT, IT and ME.)								
Course Objectives: The objective of the course is to impart:								
1.	Overall view on natural resources.							
2.	Awareness on ecosystem and its services.							
3.	Various environmental challenges induced due to unplanned anthropogenic activities.							
4.	Consciousness on the social issues, environmental legislation and global treaties							
Course Outcomes: At the end of the course, the students will be able to								
S. No	Outcome							Knowledge Level
1	Describe natural resources and their interaction							K2
2	Illustrate ecosystem types, biodiversity and conservation strategies							K2
3	Summarize contaminants of environment and preventive methods							K2
4	Explain protection of environment by employing constitutional provisions							K2
5	Explain global scenario of surroundings and social conditions							K2
SYLLABUS								
UNIT-I (8 Hrs)	Multidisciplinary Nature of Environmental Studies: Definition, Scope and Importance – Need for Public Awareness. Natural Resources: Renewable and non-renewable resources – Natural resources and associated problems. Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources							
UNIT-II (8 Hrs)	Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem. b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries) Biodiversity and Its Conservation : Introduction and Definition - genetic, species and							

	ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.
UNIT-III (6 Hrs)	Environmental Pollution: Definition, Cause, effects and control measures of: a. Air Pollution. b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards. Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides
UNIT-IV (6 Hrs)	Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy watershed management – Resettlement and rehabilitation of people; its problems and concerns – Carbon credits, Mission LiFE. Environmental ethics: Issues and possible solutions. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust –Wasteland reclamation – Consumerism and waste products. Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act. Issues involved in enforcement of environmental legislation – Public awareness.
UNIT-V (6 Hrs)	Human Population And The Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies. Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.
Text Books:	
1.	Erach Bharucha, Text book of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2019.
2.	Palaniswamy, Environmental Studies, 2/e, Pearson education, 2014.
3.	S. Azeem Unnisa, Environmental Studies, Academic Publishing Company, 2021.
4.	K. Raghavan Nambiar, “Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus”, SciTech Publications (India), Pvt. Ltd, 2010.
5.	K. V. S. G. Murali Krishna, The Book of Environmental Studies, Savera Publishing House.
6.	Environmental Studies, R. Rajagopalan, 2 nd Edition, 2011, Oxford University Press.

Reference Books:	
1.	Deeksha Dave and S.S. Katewa, Textbook of Environmental Studies, 2/e, Cengage Learning.
2.	M. Anji Reddy, “Textbook of Environmental Sciences and Technology”, BS Publication, 2014.
3.	J.P. Sharma, Comprehensive Environmental studies, Laxmi publications, 2006.
4.	J. Glynn Henry and Gary W. Heinke, Environmental Sciences and Engineering, Prentice Hall of India Private limited, 1988.
5.	G.R. Chatwal, A Text Book of Environmental Studies, Himalaya Publishing House, 2018
6.	Gilbert M. Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, 1/e, Prentice Hall of India Private limited, 1991.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc23_hs155/preview
2.	https://www.edx.org/learn/environmental-science/rice-university-ap-r-environmental-science-part-3-pollution-and-resources?index=product&objectID=course-3a6da9f2-lec07.pdf (iasri.res.in)
3.	http://ecoursesonline.iasri.res.in/Courses/Environmental%20Science-I/Data%20Files/pdf/lec07.pdf
4.	https://www.youtube.com/watch?v=5QxxaVfgQ3k

